



Smart ID Locator Using Sensor and Mobile Technology

PRODUCT REQUIREMENTS DOCUMENT (PRD)

PROJECT OVERVIEW

Smart ID Locator Using Sensor and Mobile Technology is a smart identification system designed to help students locate a misplaced or missing school ID. The system embeds a coin-sized Bluetooth Low Energy (BLE) sensor module inside a standard school ID card and connects it to a companion mobile application.

The mobile application allows students to monitor the proximity of their ID, receive alerts when the ID is separated from their phone, verify ownership, and view the last known location of the ID. The system may also provide access to the school's security office to assist in the recovery and verification of lost IDs.

PROBLEM STATEMENT

Students often misplace or lose their school IDs in classrooms, canteens, and other areas. This can result in replacement costs and delays in getting a new ID. Lost IDs may also be misused by other people to impersonate the owner. Current school IDs do not have a way to track their location or verify their rightful owner.

PROPOSED SOLUTION

The Smart ID Locator uses a BLE sensor embedded inside the school ID. A mobile application connects to the sensor to track the ID's proximity. The system sends alerts when the ID separates from the owner's phone and provides owner verification. It also records the last known GPS location to help students and security personnel find the missing ID.

TARGET USERS

Students – Students who use the Smart ID Locator to track and locate their school ID.

Faculty and Staff – School personnel who use school IDs and may use the system to help locate their IDs.

CORE FEATURE LIST

FEATURES	DESCRIPTION
BLE Proximity Tracking	Uses a low-power Bluetooth Low Energy (BLE) sensor embedded in the school ID to monitor its proximity to the user's mobile phone.
Loss Alerts	Sends an alert when the school ID moves out of BLE range from the owner's mobile phone.
Owner Verification	Uses photo matching to confirm the registered owner of the school ID.
GPS Last-Seen Logging	Records and displays the last known GPS location of the school ID to help users locate a missing ID.

USER PERSONAS



STUDENT

To easily locate their school ID when it is misplaced and quickly recover it before it becomes a bigger problem.

Needs

- BLE proximity tracking
- Loss alerts
- GPS last-seen location
- Difficulty locating a misplaced ID



FACULTY/ STAFF

Faculty and staff are school ID users who may misplace their IDs and need a convenient way to keep track of them.

Needs

- BLE proximity tracking
- Loss alerts
- GPS last-seen location
- Difficulty locating a misplaced ID

USER STORIES

User 01

Register School ID

As a student, I want to register my school ID in the mobile application, so that my ID can be connected to my account.

User 05

Monitor ID Connection

As a faculty or staff member, I want to monitor the connection between my ID and phone, so that I can know when the connection becomes weak or unavailable.

User 02

Connect ID Sensor

As a student, I want to connect my ID's BLE sensor to my phone, so that I can monitor my ID's proximity.

User 06

Check Signal Status

As a faculty or staff member, I want to view my ID's signal status, so that I can be aware of changes in its proximity.

User 03

Check ID Proximity

As a student, I want to check the proximity of my school ID, so that I can know whether my ID is nearby.

User 07

Track Last Detection

As a faculty or staff member, I want to check where my ID was last detected, so that I can identify the area where I may have misplaced it.

User 04

Receive Loss Alert

As a student, I want to receive an alert when my ID moves out of range, so that I can notice when my ID becomes separated from me.

User 8

Receive Warning

As a faculty or staff member, I want to receive a warning when my ID reaches a weak connection range, so that I can check my ID before losing contact with it.

SYSTEM ARCHITECTURE DIAGRAM

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